# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year: 2024
Programme: BCSIT | Full Marks: 100
Course: Mathematics II | Pass Marks: 45
Time: 3 hrs.

*Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.*

Section "A"

**Very Short Answer Questions**

Attempt all the questions. [10×2]

1. Use De Mover's theorem, to find the value of $[3(cos15° + isin15°)]^6$
2. Test the convergence of the series $\sum \frac{n+1}{n^3+3}$ by Limit Comparison Test.
3. Integrate $\int_1^{\infty} x^{-2} dx$
4. If f(x) = $x^3+3x^2y+3xy^2+y^3$, find partial derivative with respect to x and partial derivative with respect to y.
5. Find the general solution of $\frac{dy}{dx} = \frac{y}{x+1}$.
6. Write the definition of prime and state the fundamental theorem of arithmetic.
7. Find the period of the function f(x) = cosnx
8. If x – i y = $\frac{a-ib}{a+ib}$ then prove that $x^2 + y^2$ = 1.
9. Find the arc length of the function $y = \frac{2}{3}x^{3/2}, 0 \leq x \leq 1$
10. If u = $2x + y^3 - 3x^2y$, the show that $u_{xx} + u_{yy} = 0$.

Section "B"

**Descriptive Answer Questions**

Attempt **any five** questions. [5×10]

11. a. Find the square root of -5 +12i.

    b. If $u = log\sqrt{x^2 + y^2 + z^2}$ , the prove that

$$u_{xx} + u_{yy} + u_{zz} = \frac{1}{x^2 + y^2 + z^2}$$

12. a. Test the convergence of series by ratio test.

$$\frac{1}{1+3} + \frac{2}{1+3^2} + \frac{3}{1+3^3} + \ldots\ldots$$

    b. Find the volume of the surface generated by revolving the region bounded by the curve $y = \sqrt{x}$ and lines $y = 2$ and $x = 0$ about the line $y = 2$.

13. a. A researcher at a college of agriculture estimated that annual profit at a local firm can be described by the function

$$P = 1600x + 2400y - 2x^2 - 4y^2 - 4xy,$$

Where P equals annual profit in dollars, x equals the number of acres planted with soybeans, and y equals the number of acres planted with corn. Determine the number of acres of each crop that should be planted if the objective is to maximize the annual profit. What is the expected maximum profit?

b. Divide the polynomial $x^4 + 2x^2 + 17x - 48$ by $x + 3$ using long division. : Verify the answer by the division algorithm.

14. Solve **any two**

a. $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

b. $\tan x \frac{dy}{dx} + y = \sec x$

c. Solve the differential equations $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^{5x}$

15. a. Find the Fourier series of the function $f(x) = x^2$ for $-\pi \le x \le \pi$.

b. Find the half range cosine series of $f(x) = \pi - x$ ; $0 \le x \le \pi$

16. a. If $a, b, m \in Z$ with $m > 0$. Then show that $a \equiv b mod(m)$ if and only if $a(mod\ m) = b(mod\ m)$.

b. Find the general and particular solution of the differential equation $\frac{d^2y}{dx^2} = 2x - 9,\ f'(2) = 10, f(-2) = -10.$

Section "C"

## Long Answer Question

Attempt **any two** questions. [2×15]

17. a. Find the roots of the equation $Z^5 + 1 = 0$, Using De-moivres theorem. [7]

b. The Series $\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \ldots + \frac{1}{n^p}$ [8]

i. Converges if p>1

ii. Diverges if p≤1

18. a. Show that $\int_0^{\pi/2} sin^4\theta . cos^6\theta . d\theta = \frac{3\pi}{512}$ [8]

b. Find the Fourier series of the function [7]

$$f(x) = \begin{cases} \pi + x & if \quad -\pi < x < 0 \\ 0 & if \quad 0 \le x < \pi \end{cases}$$

19. a. A farmer can produce $f(x,y) = 200\sqrt{6x^2 + y^2}$ units of product

by utilizing x units of labor and y units of capital. [7]

i. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

ii. Evaluate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for x=10 and x=5 and interpret the result.

b. If a+ib = $\sqrt{\frac{1+i}{1-i}}$, Prove that $a^2+b^2=1$ [4]

c. Show that [4]

$$\beta(m,n)\beta(m+n,l) = \beta(n,l)\beta(n+l,m)$$